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100/100

Nano-Materials for Sustainable Energy (ChE 494)

Quiz 3: Principles of Solar Energy Harvesting

April 23, 2020 **40 Minutes** **Total: 100**

1. Sunlight is a form of "electromagnetic radiation" and the visible light that we see is a small subset of the electromagnetic spectrum.

   (A) True (B) False

2. Light is a type of quantum-mechanical particle called a/an ....

   (A) Electron (B) Photon (C) Phonon (D) Exciton

3. What is the relationship between the energy of a photon (E) and the wavelength of the light (λ)? Inverse relationship between E and λ

   $E = \frac{hc}{\lambda}$ where h = Planck's constant $\equiv 6.626 \times 10^{-34}$ J·s
   c = speed of light $\equiv 2.998 \times 10^{8}$ m/s

4. What is photon flux?

   photon flux, $\phi = \frac{\text{\# of photons}}{\text{area} \times \text{time}}$

5. The Air Mass is the path length which light takes through the atmosphere normalized to the shortest possible path length (that is, when the sun is directly overhead).

   (A) True (B) False

6. Semiconductors act as ~~conductors~~ insulators at low temperatures and ~~insulators~~ conductors at higher temperatures.

   (A) True (B) False

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7. The band gap of a semiconductor is the minimum energy required to excite an electron that is stuck in its bound state into a free state where it can participate in conduction.

(A) True (B) False

8. Doping creates N-type material when semiconductor materials from group IV are doped with .... atoms.

(A) Group III (B) Group IV (C) Group V (D) Group VI

9. What is law of mas action?

$n_o p_o = n_i^2$ At equilibrium, the product of the majority and minority carrier conc. is a constant.

where

n_i = intrinsic carrier conc.

n_o = equilibrium electron conc.

p_o = equilibrium hole carrier conc.

10. When the energy of a photon is equal to or greater than the band gap of the material, the photon is absorbed by the material and excites an electron into the conduction band.

(A) True (B) False

11. What is radiative (band-to-band) recombination?

Radiative recombination is the recombination mechanism that dominates in direct bandgap semi-conductors. An electrode from the conduction band directly combines with a hole in the valence band & releases a photon. Emitted photon has energy similar to band gap & is only weakly absorbed such that it can exit the piece of semiconductor.

12. Auger Recombination involves .... carriers.

(A) one (B) three (C) two (D) four

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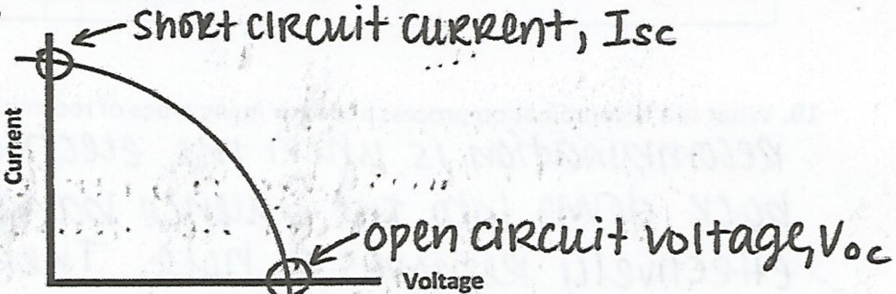
13. Auger recombination is most important at high carrier concentrations caused by heavy doping or high-level injection under concentrated sunlight.

(A) True **(B) False**

14. Write the current-voltage (I-V) relationship for a P-N junction diode.

$$I = I_o\left(e^{qV/kTn} - 1\right)$$

15. The I-V curve of a solar cell is shown below. Indicate the short-circuit current density and open-circuit voltage?



16. Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun. Write the efficiency of a solar cell in terms of short-circuit current density, open-circuit voltage, fill factor, and input solar energy?

$$\text{Efficiency, } \eta = \frac{V_{oc} I_{sc} FF}{P_{in}}$$

17. What is the difference between direct and indirect bandgap semiconductors?

Direct band gap semiconductors have the top of the valence band & the bottom of the conduction band occur at the same value of momentum. In indirect band gap semiconductors, the maximum energy of the valence band occurs at a different value of momentum as the minimum in the conduction band energy.

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18. Fill-out the following table for N-type and P-type silicon semiconductor?

	N-Type Silicon	P-Type Silicon
Dopant	Group V (e.g. phosphorous)	Group III (e.g. Boron)
Majority Carriers	Electrons	Holes
Minority Carriers	Holes	Electrons

19. What is a recombination process and how many types of recombination process are there?

Recombination is when the electron stabilizes back down into the valence band & thus also effectively removes a hole. There are 3 basic types of recombination: radiative, Auger, & Shockley-Read-Hall.

20. The absorption coefficient determines how far into a material light of a particular wavelength can penetrate before it is absorbed?

(A) True

(B) False